

BRISTOL, United Kingdom

WMO: 037243

Lat: 51.383N

Lon: 2.719W

Elev: 622

StdP: 14.37

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.3	28.4	21.3	16.3	29.3	23.4	18.1	30.5	32.8	47.1	29.8	46.0	10.3	60	0.647

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	76.6	64.1	73.0	62.7	69.7	61.4	65.6	72.6	63.9	69.9	62.6	67.6	9.9	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.0	88.1	67.0	62.3	85.8	66.2	60.7	81.2	64.6	30.9	72.7	29.7	70.6	28.6	66.8	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.6	23.5	21.1	DB	22.8	81.8	3.2	4.1	20.5	84.8	18.6	87.2	16.8	89.5	14.5	92.5	
			WB	22.0	69.0	3.2	2.4	19.7	70.7	17.9	72.1	16.1	73.5	13.8	75.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.1	40.4	40.7	43.5	47.5	52.4	57.9	61.3	60.7	57.0	51.8	45.6	41.5
	DBStd	8.97	5.70	5.40	5.04	4.88	4.80	4.38	4.22	3.74	3.82	4.54	5.21	6.19
	HDD50	1342	299	261	205	103	27	1	0	0	1	32	145	268
	HDD65	5485	763	680	666	526	390	219	133	145	242	411	581	729
	CDD50	1366	1	1	4	27	102	237	349	331	210	87	14	3
	CDD65	33	0	0	0	0	0	5	17	10	1	0	0	0
	CDH74	217	0	0	0	0	3	46	113	50	4	1	0	0
	CDH80	26	0	0	0	0	0	6	13	7	0	0	0	0
Wind	WSAvg	10.9	12.4	12.1	11.9	10.9	10.6	10.1	9.8	9.5	9.8	10.9	10.9	11.8
Precipitation	PrecAvg	37.2	3.8	2.5	2.9	2.3	2.6	2.7	2.7	3.1	3.5	3.5	3.7	4.0
	PrecMax	45.9	7.0	5.8	7.2	4.4	7.1	6.4	6.5	6.7	8.3	7.8	8.1	8.6
	PrecMin	25.9	0.7	0.1	0.2	0.2	0.3	0.4	0.6	0.7	0.8	0.3	1.7	0.8
	PrecStd	5.0	1.8	1.6	1.5	1.1	1.7	1.5	1.5	1.7	1.9	2.0	1.7	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.7	55.2	59.4	68.3	73.3	80.3	82.1	80.2	73.6	66.0	58.7	55.4
		MCWB	51.6	50.3	51.1	56.4	61.1	63.7	65.5	65.3	63.3	60.8	57.2	53.1
	2%	DB	51.8	51.6	55.2	62.4	68.2	73.5	77.2	73.6	68.4	61.2	55.8	53.6
		MCWB	50.1	49.2	49.6	53.6	59.0	62.1	64.1	64.5	61.2	58.3	54.1	52.2
	5%	DB	50.0	49.8	52.1	57.6	64.4	69.8	73.2	69.9	65.9	60.4	55.0	51.9
		MCWB	49.0	48.1	48.3	50.7	56.8	60.6	62.8	62.5	59.6	57.6	53.4	50.4
	10%	DB	48.3	48.2	50.3	55.2	60.8	66.1	69.6	67.6	63.0	58.7	53.4	50.1
		MCWB	47.6	47.0	47.7	49.5	55.1	58.4	61.4	61.3	58.4	56.2	51.6	49.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.4	51.9	52.9	57.4	63.6	67.1	67.5	67.7	65.2	62.2	57.5	53.9
		MCDB	53.5	54.1	56.9	65.2	69.4	75.8	77.0	74.3	70.5	65.0	58.5	54.7
	2%	WB	50.4	49.7	51.2	54.1	60.2	63.3	65.5	65.5	62.8	59.6	55.4	52.7
		MCDB	51.2	50.8	54.0	60.9	66.2	70.9	74.0	70.8	66.8	61.5	56.3	53.6
	5%	WB	48.6	48.2	49.8	52.2	57.8	61.3	63.7	63.7	61.1	58.2	53.6	51.0
		MCDB	49.1	48.9	51.5	56.7	63.0	67.8	70.7	68.4	64.4	59.9	54.7	52.0
	10%	WB	47.3	46.8	48.3	50.4	55.7	59.6	62.1	62.3	59.7	56.7	52.0	49.2
		MCDB	47.8	47.7	49.9	53.7	60.1	64.7	67.6	66.2	62.6	58.4	53.2	49.9
Mean Daily Temperature Range	5% DB	MDBR	7.8	8.5	10.3	12.3	12.8	13.2	12.9	11.9	11.6	9.4	8.0	7.4
		MCDBR	8.8	9.1	12.7	16.7	18.0	19.6	19.7	16.0	14.9	10.0	8.1	7.9
		MCWBR	9.1	7.7	9.1	10.2	10.8	10.8	9.5	8.6	8.9	7.5	7.5	8.3
	5% WB	MCDBR	8.5	8.4	10.8	14.4	15.8	16.7	16.3	13.7	12.4	9.1	7.6	7.9
		MCWBR	9.1	8.2	8.7	9.6	10.2	10.0	9.2	8.3	8.9	7.9	7.7	8.5
Clear-Sky Solar Irradiance	taub		0.321	0.335	0.366	0.384	0.389	0.382	0.379	0.367	0.358	0.345	0.320	0.310
	taud		2.375	2.400	2.313	2.275	2.298	2.344	2.356	2.408	2.429	2.444	2.423	2.351
	Ebn at Noon		219	250	262	270	274	277	276	273	263	243	218	201
	Edh at Noon		20	25	33	38	39	38	36	33	29	24	19	17
All-Sky Solar Radiation	RadAvg		249	471	835	1288	1580	1715	1607	1333	1004	568	302	194
	RadStd		23	57	107	143	80	180	148	116	73	49	29	24

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (70 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page